

EE366200 DSP Laboratory

Lab 2 Audio & Speech: invMFCC

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1. Introduction

Mel-Frequency Cepstral Coefficients (MFCCs) are a compact feature representation designed to model the way humans perceive sound. They are derived by transforming the short-time spectrum of a signal through mel-scaled filter banks, logarithmic compression, and a discrete cosine transform, thereby emphasizing perceptually meaningful spectral patterns while reducing dimensionality. Because MFCCs are so effective at capturing spectral envelopes, they have become a standard representation for speech recognition, speaker identification, and other audio classification tasks. However, this dimension reduction and nonlinear transformation inevitably discard or distort parts of the original signal. This raises an important question: given a set of MFCCs, how well can we reconstruct the original time-domain waveform?

The second laboratory session is devoted to exploring the process of **MFCC inversion**. In this exercise, the extracted cepstral features are first mapped back to the log-mel domain by applying the inverse discrete cosine transform. The exponential function is then used to restore linear-scale mel-band energies. To

approximate the short-time spectrum, these mel energies are inverted through least-squares synthesis against the triangular mel filterbank, effectively solving an ill-posed inverse problem. Once a magnitude spectrum is obtained, the final step is to recover the missing phase information and reconstruct the waveform through iterative algorithms such as the Fast Griffin–Lim Algorithm (FGLA), which enforces both magnitude constraints and short-time Fourier transform consistency.

By carrying out this reconstruction pipeline, the goal of Lab 2 is to deepen understanding of which components in the MFCC extraction process are invertible, which introduces irreversible loss, and how these limitations manifest as audible distortions in the reconstructed signal. Moreover, the experiments allow us to observe the influence of design choices such as the number of mel filter banks, frame parameters, and preprocessing steps on the reconstruction quality. Through both theoretical analysis and practical implementation, this lab highlights the challenges of inverting a perceptually motivated representation back into the time domain, setting the stage for further discussion of artifacts, parameter effects, and possible algorithmic improvements.

2. Objectives

The purpose of this laboratory is to examine how a time-domain speech signal can be reconstructed from its Mel-Frequency Cepstral Coefficients and to explore

the inherent limitations of this inversion process. Since MFCCs are designed primarily as compact descriptors for recognition tasks rather than for reconstruction, this experiment aims to determine to what extent the original waveform can be recovered and what distortions inevitably arise. Particular emphasis is placed on analyzing the invertibility of the MFCC extraction pipeline by tracing each stage in reverse, including the inverse discrete cosine transform, the restoration from log to linear scale, the approximation of the short-time spectrum through mel filterbank inversion, and the recovery of phase information using the Fast Griffin–Lim Algorithm. By conducting this series of experiments, the lab seeks to provide insights into which transformations preserve sufficient information for reconstruction, which introduce irreversible loss, and how design choices such as filterbank configuration and frame parameters influence the perceptual quality of the reconstructed signal.

3. Approach

The reconstruction experiment begins with the analysis of the speech waveform in the frequency domain. A short-time Fourier transform (STFT) is first applied to the signal, producing a spectrogram that captures the temporal evolution of its frequency components. Two mel filterbanks with different resolutions are then constructed: one with 12 bands and the other with 64 bands. Each filterbank is

applied to the STFT magnitude to obtain mel-band energies, which are then passed through a logarithmic scaling and a discrete cosine transform (DCT) to produce the mel-frequency cepstral coefficients (MFCCs). To directly observe the effect of filterbank resolution, a single analysis frame is randomly selected from the STFT, and its MFCCs are computed under both configurations. The coefficients from the 12-band and 64-band cases are then plotted side by side for comparison. This result constitutes Demo Checkpoint 1 and is presented in Figure 1, where the two sets of coefficients are shown for the same frame.

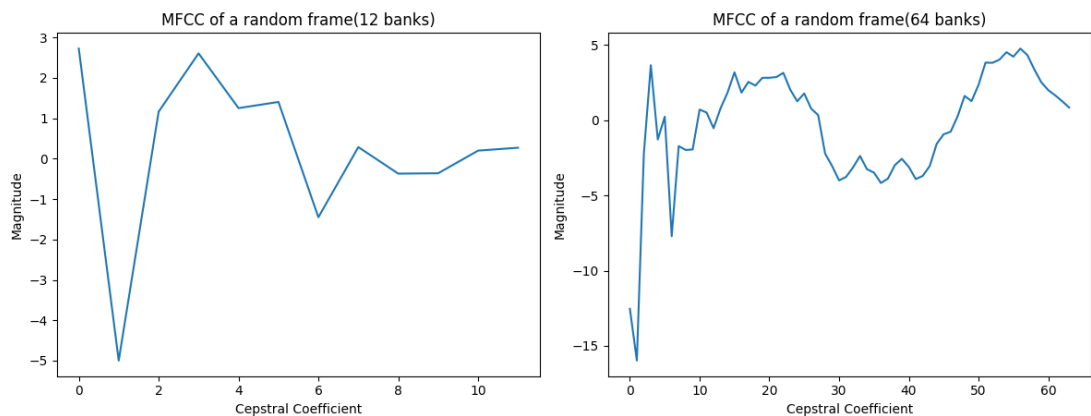


Figure1. Random frame of the MFCC using the Mel-filter with 12 banks(left) and 64 banks(right)

After the single-frame comparison, the full sequences of MFCCs across time are visualized for both filterbank settings. The coefficients are displayed as heatmaps, providing a clear view of how the choice of filterbank resolution affects the temporal and cepstral patterns. This constitutes Demo Checkpoint 2, with the results arranged in Figure 2: the left panel shows the MFCC heatmap for the 12-band

configuration, while the right panel shows the MFCC heatmap for the 64-band configuration, both plotted with identical color scales for direct comparison.

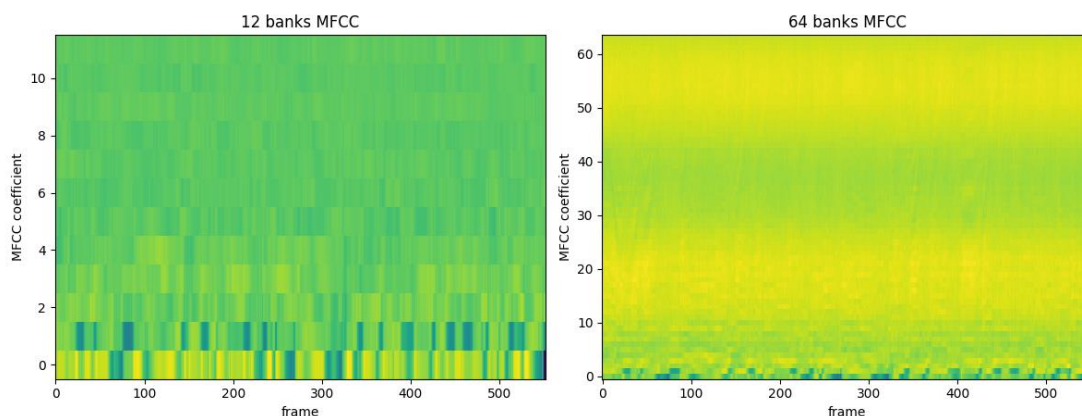


Figure2. Heatmap of the MFCC using the Mel-filter with 12 banks(left) and 64 banks(right)

The third checkpoint examines the inversion process, where time-domain signals are reconstructed from MFCCs. This procedure involves applying the inverse discrete cosine transform (invDCT) to recover log-mel energies, exponentiating to restore linear magnitudes, performing least-squares inversion of the mel filterbank transformation to estimate the STFT magnitude, and finally reconstructing the waveform with the Fast Griffin–Lim Algorithm (FGLA), which iteratively recovers consistent phase information. The original signal and the reconstructed signals obtained from the 64-band MFCCs are plotted in Figure3, which forms Demo Checkpoint 3, illustrating the difference between these two spectrogram.

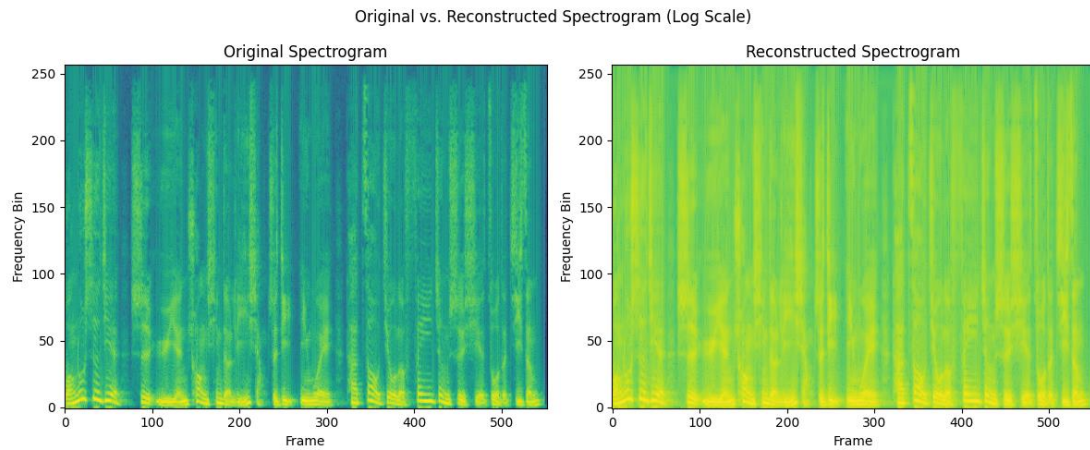


Figure3. The original spectrogram(left) and the reconstructed one(right)

Finally, the fourth checkpoint extends the reconstruction experiment to include the effect of pre-emphasis. Four sets of waveforms are generated, corresponding to the combinations of 12-band and 64-band MFCCs with and without pre-emphasis. These reconstructed audio files form Demo Checkpoint 4. Although the .wav files are not included directly in the report, they are systematically produced following the same inversion pipeline and are submitted as part of the lab assignment for further evaluation and discussion.

4. Discussion

In this section, we provide reflections on the MFCC extraction pipeline and the process of reconstruction by addressing the questions given. The following analysis connects the experimental results with both intuitive explanations and mathematical formulations.

Lecture Question 1:

Why not using rectangular filters for “energy” calculation? We avoid rectangular passbands when computing mel “energies” because hard-edged, non-tapered filters make the band energy extremely sensitive to tiny shifts of spectral peaks across the band boundary, producing blocky, discontinuous trajectories that neither reflect cochlear smoothing nor remain stable when the spectrum slightly moves from frame to frame; in contrast, the standard overlapping triangular mel filters provide smooth, approximately constant-Q weighting that better mimics auditory frequency resolution and yields band energies that vary continuously with frequency and time, which is exactly what we want before taking the log and DCT in MFCCs. The triangular, overlapping design also acts like a linear interpolation over neighboring FFT bins, so when we later invert mel energies (e.g., via least-squares) the problem is better conditioned than with rectangular “bucket” aggregation that throws away within-band structure and leaves sharp steps that are difficult to reconstruct without artifacts. In short, rectangular filters over-quantize the spectrum and amplify boundary effects, whereas triangular overlapping filters smooth, interpolate, and align with perceptual spacing—hence they are preferred for mel-energy computation in MFCC pipelines.

Lecture Question 2:

Why should mel-filters overlap? Mel filters are designed with overlapping triangular bands because this structure better matches the way the human auditory system processes sound, where frequency analysis occurs in partly overlapping critical bands rather than in isolated ranges. Overlap ensures that no spectral gaps are left between adjacent filters, so the entire frequency axis is smoothly covered and the transition of energy between neighboring bands remains continuous. This smoothness prevents abrupt changes in mel energies when formants or harmonics shift slightly, thereby providing a more stable and perceptually relevant description of the spectral envelope. Furthermore, because the outputs of overlapping filters are correlated, the subsequent discrete cosine transform can effectively de-correlate them and concentrate most of the meaningful information into the low-order cepstral coefficients, which is the key purpose of MFCC representation. Finally, overlapping filters reduce the sensitivity to discretization and windowing effects in the STFT, which helps both in feature extraction and in later reconstruction tasks.

Lecture Question 3:

Why are high-quefreny MFCCs usually abandoned when doing speech recognition? Because the discrete-cosine transform (DCT) orders cepstral information by “quefreny,” the low-quefreny MFCCs compactly describe the

slowly varying spectral envelope (formants) that carries phonetic content, while the high-frequency terms mainly encode rapid spectral ripples—pitch harmonics, fine excitation details, and channel/microphone ripples—which fluctuate strongly across speakers and recording conditions; in other words, they have low task-relevance and low robustness for recognition, yet they increase feature dimensionality and sensitivity to noise and misalignment. Since MFCC is a *dimension-reduction* front end designed to summarize log-mel energies, most discriminative power concentrates in the first few coefficients, so it is better to keep only the lower cepstra (often with cepstral liftering) and discard the high-frequency coefficients to improve invariance and generalization.

Lecture Question 4:

Do you think MFCC is good for speaker identification purposes? How about baby-cry detection? — Yes, to both. MFCC (log-mel filterbank energies followed by DCT on pre-emphasized, short-time spectra) primarily encodes the spectral envelope of speech, which is shaped by the speaker's vocal-tract configuration. Those anatomical/acoustic traits are speaker specific yet relatively stable across phonetic content, so MFCCs provide a compact, discriminative representation for speaker identification when aggregated over frames. The pre-emphasis step flattens the spectral slope and brings out high-frequency cues that further help separate

speakers. For baby-cry detection, MFCCs are also effective because cries exhibit stereotyped spectral shapes—strong low-to-mid-band energy with harmonic structure and slowly varying formant-like peaks—which MFCCs capture well through the mel-scaled filtering and log compression; using the same pre-emphasized front end enhances the visibility of those envelope patterns, making MFCCs a pragmatic and reliable choice for detecting infant cries.

Lecture Question 5:

If two sounds have similar MFCCs, does that imply they sound similar to our ears? In short, similar MFCCs suggest similarity mainly in the short-time spectral envelope on the mel scale (i.e., the overall timbre/formant shape and energy distribution), but they do not guarantee overall perceptual similarity. MFCCs are obtained by mapping a frame's magnitude spectrum to mel bands, taking logs, and then applying a DCT, so closeness in this feature space primarily reflects closeness of the log-mel spectral envelope rather than all properties of the waveform. As emphasized in the lectures, MFCCs are a dimension-reduced representation of DCT (log-mel-energies), so some information is intentionally discarded; thus, two signals with near-by MFCC vectors can still differ in aspects listeners notice, such as pitch, transient detail, and phase structure.

Concretely, if two signals have similar MFCC features, the part that is most alike is the filter-like component of the sound: the broad, slowly varying spectral envelope associated with the vocal-tract or instrument resonances. This is exactly what MFCCs are designed to capture in short time: the smooth spectral shape that best characterizes “what kind of sound” it is, rather than the fine harmonic spacing or instantaneous phase. In the source–filter view of speech, this means two utterances with close MFCCs tend to share a similar vocal-tract (filter) response even if their source differs.

On the other hand, perceptual similarity depends on more than the envelope alone. MFCCs do not encode phases and largely suppress fine harmonic structure and rapid temporal events; as a result, signals with different fundamentals (pitch), different transient onsets, or different phase characteristics may still yield similar MFCCs yet sound different. The inversion lab further illustrates this limitation: when reconstructing from MFCCs, we can approximate magnitudes but must recover phase iteratively, and artifacts appear precisely because information outside the envelope is missing—underscoring that MFCC proximity is not a sufficient condition for perceptual sameness.

Lecture Question 6:

Is $(WW^t)^{-1}y$ guaranteed to be non-negative given that all entries in y are non-negative? In general, the answer is no, and the reason can be shown both algebraically and by connecting to the physical meaning of overlapping mel filters.

Let $W \in \mathbb{R}^{L \times M}$ denote the mel filterbank matrix whose rows are non-negative, overlapping triangular weights over linear-frequency bins. The Gram matrix $A = WW^t$ is symmetric positive definite, but because neighboring triangles overlap, its off-diagonal entries are strictly positive. Already in the $L = 2$ case we can write:

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \text{ with } a > 0, b > 0, c > 0$$

and

$$\det A = ac - b^2 > 0$$

Its inverse is:

$$A^{-1} = \frac{1}{ac - b^2} \begin{bmatrix} c & -b \\ -b & a \end{bmatrix},$$

which has negative off-diagonals. Choosing a perfectly valid non-negative vector

$y = [1, 0]^t$ gives $A^{-1}y = \frac{1}{ac - b^2} [c, -b]^t$, whose second component is negative.

Hence even if all mel-band energies in y are non-negative, the vector $(WW^t)^{-1}y$ generally contains mixed signs; algebraically this is unavoidable whenever A has positive off-diagonals, and physically it reflects that inverting an overlapping smoothing requires subtracting neighbors to “unmix” correlated bands.

Having established this for $(WW^t)^{-1}y$, we can examine the expanded back-projection $x = W^t(WW^t)^{-1}y$ used to synthesize linear-frequency magnitudes. The extra multiplication by $W^t \geq 0$ does not restore elementwise non-negativity, because it forms weighted sums of the entries of $(WW^t)^{-1}y$, and those entries already have mixed signs. A concrete triangular filterbank makes this explicit. Let two overlapping triangles live on three frequency bins so that:

$$W = \begin{bmatrix} 1 & 0.5 & 0 \\ 0 & 0.5 & 1 \end{bmatrix}, A = WW^t = \begin{bmatrix} 1.25 & 0.25 \\ 0.25 & 1.25 \end{bmatrix}, A^{-1} = \frac{1}{1.5} \begin{bmatrix} 1.25 & -0.25 \\ -0.25 & 1.25 \end{bmatrix}$$

With $y = [1, 0]^t$ we obtain $z = A^{-1}y = [0.8\bar{3}, -0.1\bar{6}]^t$, and then:

$$\hat{x} = W^t z = \begin{bmatrix} 1 & 0 \\ 0.5 & 0.5 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0.8\bar{3} \\ -0.1\bar{6} \end{bmatrix} = \begin{bmatrix} 0.8\bar{3} \\ 0.\bar{3} \\ -0.1\bar{6} \end{bmatrix},$$

so a linear-frequency bin remains negative despite the non-negativity of both W and y . Interpreted physically, mel filtering is a many-to-one, overlapping averaging of spectral bins; its algebraic inverse must “undo” overlap by subtracting neighboring contributions, and those subtractions manifest as negative components either before or after multiplying by W^t . Finally, these negative values do not violate any energetic principle in the reconstruction program because the time-domain synthesis is driven by magnitudes within the Griffin–Lim loop. When a back-projected bin happens to be negative, it is simply $-|x_k|$, which is equivalent to the positive magnitude $|x_k|$ with an added phase of π . The algorithm enforces the target magnitude at every iteration and lets the phase absorb the sign, so the

complex STFT used for inverse STFT has non-negative magnitudes by construction while any sign from the linear algebra becomes a harmless phase flip. In this way, the mathematical fact that $(WW^t)^{-1}y$ and $W^t(WW^t)^{-1}y$ are not guaranteed to be non-negative is fully consistent with the physical picture of overlapping filters and is gracefully handled during phase-aware waveform reconstruction.

Report Question 1:

What are the artifacts and distortions in the reconstructed audio, and which parts of the MFCC pipeline cause them (i.e., which sections are not invertible)? In our reconstructions the most salient artifacts are a “smoothed” or muffled timbre together with blurred consonant onsets, occasional metallic/phasiness in sibilants, and a slight drift or weakening of fine harmonic lines in the spectrogram; these degradations arise even when we keep all cepstral coefficients because several operations in the MFCC pipeline either discard information or are ill-conditioned to invert. The first and dominant source is the loss of phase when we take the magnitude of the STFT: MFCCs are computed from $|STFT|$ and therefore throw away the instantaneous phase that encodes temporal microstructure, so any phase we later recover is only an estimate, which explains the metallic/phasiness and transient smear. A second, structural source is mel filtering: mapping hundreds of linear-frequency FFT bins into a much smaller number of overlapping mel bands is

a many-to-one linear transform; even with a pseudo-inverse the problem is ill-conditioned, so the spectrum we synthesize is an interpolated approximation that over-smooths formants and weakens fine harmonic ripples, producing the “muffled” quality and the smoothed spectrogram we observe. The log compression is mathematically invertible (via $\exp$), but it amplifies small numerical errors in low-energy regions when we go back to linear scale, which contributes to hissy, whisper-like high-frequency residue after inversion. DCT is unitary and thus invertible in principle; however, if one reduces dimensionality, the discarded high-frequency terms remove rapid spectral variations and make attacks and sibilants sound duller—this is why the 12-bank setting yields a noticeably smoother, less detailed reconstruction than 64 banks. In short, the artifacts are chiefly the audible footprint of (1) non-invertible phase removal at the STFT magnitude step and (2) the rank-deficient, ill-conditioned mel filterbank compression; log/exp nonlinearity and any cepstral truncation further magnify these losses, together yielding the metallic, smoothed, and transient-blurred character heard in the reconstructed audio.

Report Question 2:

For Question 2, which asks “Experiment with different frame length, step length, and number of banks; discuss what effects each of them has in the reconstruction process,” we can use Figure 4 to analyze the observed results.

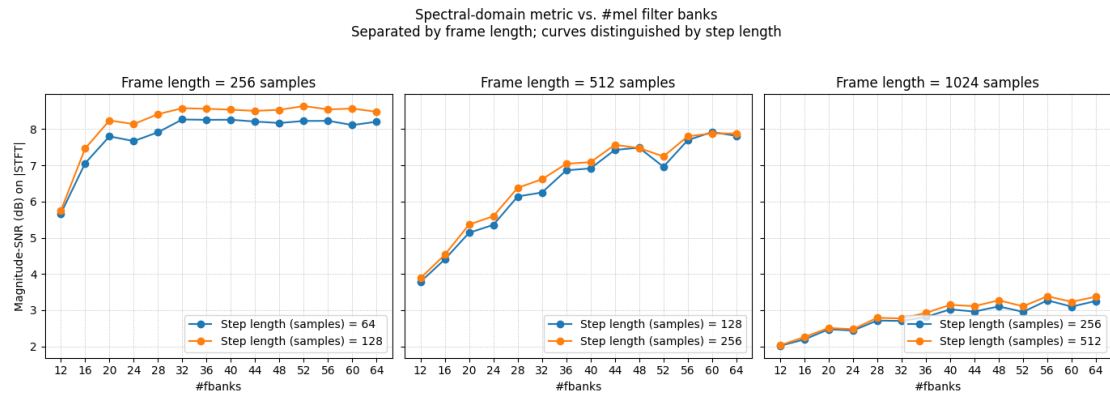


Figure4. SNR of the reconstructed STFT at different frame length

In the plots of Figure 4, the x-axis corresponds to the number of mel filter banks, while the subplots are separated by different frame lengths (256, 512, 1024 samples), and within each subplot, different step lengths are distinguished by color. From these results, one can first observe that increasing the number of fbanks consistently improves the reconstruction quality, as indicated by a monotonic rise of the magnitude-SNR curves across all subplots. This matches the intuition that a larger number of filters better captures fine-grained spectral details, thereby reducing information loss in the inversion. Looking at the effect of frame length, shorter frames (256 samples) show higher overall SNR, while longer frames (1024 samples) yield much lower values, suggesting that shorter windows provide more accurate local time-frequency representation for reconstruction. Regarding step length, smaller step sizes generally give slightly better results, particularly in the case of 256-sample frames, where the SNR curve of step length 128 is consistently

above that of 64. This indicates that overlap between frames helps enforce consistency during Griffin–Lim iterations, but when the hop is too small, redundancy may not further enhance quality. Taken together, Figure 4 demonstrates that reconstruction quality benefits most from more mel filter banks, moderately short frame lengths, and balanced step lengths, reflecting the trade-offs between spectral resolution, temporal resolution, and overlap consistency in the STFT–MFCC–inverse pipeline.

Bonus Question 1:

To answer the question, we first calculated the SNR of four signals, with each signal processed using different numbers of banks and pre-emphasis coefficient, then reconstructed by strictly following the steps in the example code.

	12 banks	64 banks
Original	4.1815(dB)	7.9683(dB)
Pre-emphasized	0.4983(dB)	0.7141(dB)

Table1. The baseline SNR of 4 different signals.

From Table 1, it's clear that the SNR of the pre-emphasized signals is significantly lower than that of the original signals. To improve the quality, we applied a de-emphasis process before calculating the SNR, compensating for the pre-emphasis applied at the beginning.

	12 banks	64 banks
Original	4.1815(dB)	7.9683(dB)
Pre-emphasized	3.8796(dB)	7.8932(dB)

Table2. The SNR of 4 different signals (with de-emphasis).

As shown in Table 2, the quality of the pre-emphasized signals improved dramatically with de-emphasis, reaching nearly the same level as the original signals. This indicates that de-emphasis is essential in recovering the quality lost during the pre-emphasis stage.

In addition to this baseline method, we also tested other reconstruction strategies, such as non-negative least squares (NNLS), Tikhonov-regularized NNLS, and KL-divergence-based NMF. However, none of these approaches produced SNR values that significantly outperformed the baseline, so their detailed results are not presented here.

Bonus Question 2:

For Bonus Question 2, we are asked to modify the DCT/IDCT sections to include dimension reduction and reconstruction, which effectively simulates compression and decompression. The idea is to retain only the first K coefficients in the DCT domain and then reconstruct the signal using the inverse transform. This

approach allows us to investigate how the reduction of coefficients influences the reconstruction quality in terms of SNR.

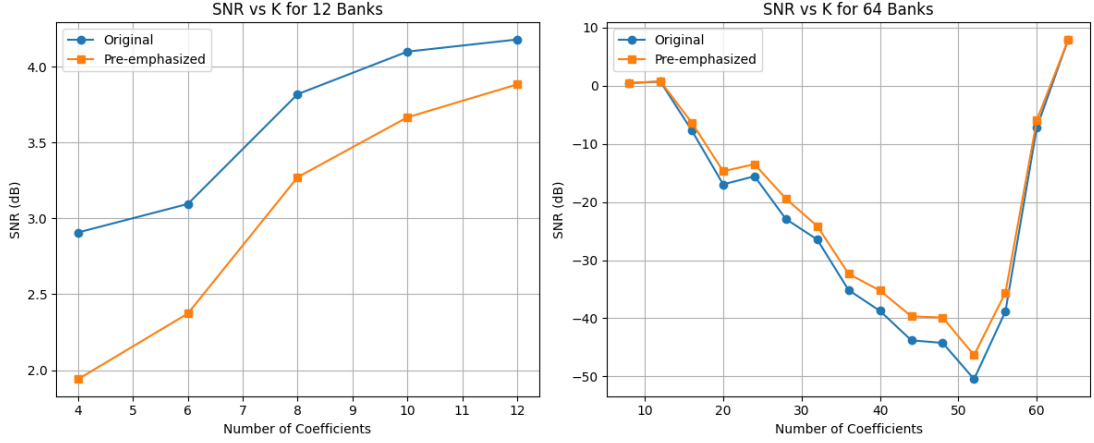


Fig5. SNR under different choice of K

As shown in Figure 5, when the number of coefficients increases, the SNR improves because more spectral information is preserved during reconstruction. For the 12-bank case, the SNR grows steadily with K and reaches its highest level when all coefficients are retained, indicating that keeping more coefficients results in better fidelity. On the other hand, for the 64-bank case, the trend is less stable: while SNR initially increases, it subsequently drops significantly in the mid-range of coefficients before recovering when nearly all coefficients are kept. This fluctuation suggests that discarding or truncating certain coefficients in a high-resolution representation may introduce severe distortions before the full set is restored.

Overall, this experiment demonstrates that the process of compression and decompression inherently reduces reconstruction quality, particularly when too few

coefficients are retained. The effect is more severe in high-bank scenarios (such as 64 banks), where the representation is more detailed and sensitive to coefficient truncation. Thus, although dimension reduction provides a way to compress the feature representation, it comes at the cost of degraded SNR, with the severity depending on both the number of banks and the number of coefficients preserved.

5. Conclusion

In this lab, we explored the inversion of Mel-Frequency Cepstral Coefficients (MFCCs) back into the time domain and examined the challenges that arise from this process. By conducting four structured checkpoints, we observed the influence of filterbank resolution on MFCC extraction, visualized how the temporal patterns of coefficients evolve, and reconstructed audio signals under different configurations. The results highlighted that while the inversion pipeline can approximate the original signal, inevitable artifacts and distortions occur due to the non-invertible nature of several MFCC extraction steps, such as the loss of phase information and the many-to-one mapping of mel filtering.

Quantitative evaluations using SNR revealed how reconstruction quality varies with frame length, step length, and the number of mel filter banks. In general, more filter banks provided higher fidelity, whereas excessively long frames degraded accuracy. Additional experiments on coefficient truncation further demonstrated

that compression leads to degraded SNR, especially in high-resolution settings, underscoring the trade-off between compactness and signal fidelity. Moreover, while alternative inversion strategies such as NNLS, Tikhonov-regularized NNLS, and NMF were tested, they did not significantly outperform the baseline pseudo-inverse method, confirming that the primary bottlenecks lie in the inherent information loss of the MFCC pipeline rather than in the linear inversion stage.

Overall, this laboratory underscored the effectiveness of MFCCs as perceptually meaningful representations, while also clarifying their limitations for waveform reconstruction. The experiments provided both visual and numerical evidence of how design choices and algorithmic assumptions impact reconstruction quality, paving the way for more advanced approaches—such as refined phase recovery methods or perceptually motivated constraints—to further improve the quality of audio regenerated from MFCCs.

6. Reference

- [1] Department of Electrical Engineering, National Tsing Hua University, *DSP_Lab_2025_Lab_1_lecture*, 2025.
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